Srinivas Likhith Raichur

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SUMMARY:

Software Engineer passionate about distributed systems and large-scale backend platforms, experienced in building fault-tolerant cloud-native applications, high-performance APIs, and resilient databases. Holds a Master's in Computer Science from Stony Brook University.

EDUCATION:

Master's in Computer Science, Stony Brook University, USA **May 2025**

Courses: Distributed Systems, Database Systems, Computational Biology, Data Visualization

Bachelor's in Computer Science, Vellore Institute of Technology, India **May 2022**

Courses: Data Structures & Algorithms, Operating Systems, Computer Networks, Object Oriented Programming

SKILLS:

Programming Languages: Java, Python, C++, C#, JavaScript, TypeScript, SQL, HTML/CSS, Shell Scripting
Frameworks & Libraries: Spring Boot, Spring MVC, ASP.NET, Django, Flask, React.js, Node.js, Nginx, REST APIs, GraphQL
Cloud Platforms & DevOps: AWS (EC2, S3, Lambda, RDS, SNS, SQS, IAM), Azure, Docker, Jenkins, Terraform, Kubernetes
Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL
Monitoring & Logging: Prometheus, Grafana, ELK Stack, Elastic APM, Datadog
Messaging & Streaming: Apache Kafka, RTMP, HLS, AWS SQS, Celery

EXPERIENCE:

Graduate Researcher | Stony Brook University **Jan 2024 – Present**

- Collected clinical datasets from Stony Brook Hospital and performed data anonymization, cleaning, and preprocessing using Python scripts to enable large-scale analysis while ensuring compliance with privacy standards.
- Designed and implemented a PostgreSQL-backed database system to store and analyze fetal heart rate (FHR) signals and clinical data, enabling large-scale querying and early disorder detection.
- Built an LLM-powered query interface using OpenAI GPT models with a FastAPI backend, converting natural language queries from healthcare professionals into optimized SQL, providing insights without requiring technical expertise.

Software Engineer | LocoNav, India **Jan 2022 – Aug 2023**

- Designed and developed backend microservices using Django that supported real-time telemetry processing and live video feeds from more than 50,000 IoT-enabled dashcams, ensuring high reliability and scalability across production environments.
- Implemented a resilient command orchestration engine using Celery, Redis, and Kafka to manage prioritized task execution with fault recovery mechanisms, ensuring consistent execution flow with retry strategies and scheduled task support.
- Architected a configuration management system using a React-based frontend interface and Django backend, which allowed administrators to manage versioned settings for device groups, eliminating manual overhead and reducing deployment time.
- Developed React-based components for the live video monitoring dashboard and integrated Nginx and HLS streaming to achieve sub-4-second latency for real-time dashcam video streams supporting multiple concurrent sessions.
- Built unit and integration tests using Django's native test framework and PyTest and integrated SonarQube into Jenkins CI/CD pipelines with custom quality gates, ensuring 85%+ code coverage and improving code reliability across releases.
- Created secure and scalable RESTful APIs with JWT authentication for vehicle tracking and on-demand video delivery using AWS S3, and used FFmpeg to overlay real-time metadata such as timestamps and object bounding boxes on video frames.
- Automated containerized deployments of Django microservices using Docker and Jenkins, enabling seamless CI/CD workflows for both staging and production environments, reducing deployment time and operational overhead.
- Integrated backend services with Elastic APM, Datadog, Prometheus, and Grafana for monitoring and performance visualization.

PROJECTS:

ConsensusX | Self-Optimizing Blockchain Platform **Feb 2024 - May 2024**

- Implemented an adaptive system integrating BFT consensus protocols (PBFT, SBFT, HotStuff) with blockchain architectures (OX, OXII, XOV, XOV++) to support varying workloads and Byzantine fault scenarios, enabling resilient and scalable transaction management.
- Built a reinforcement learning agent to optimize configuration selection, achieving ~27% higher throughput than fixed setups.

FleetUpdater - *LocoNav* | IoT Device Management **May 2023 - June 2023**

- Built a backend microservice with REST APIs for devices and ReactJS frontend, enabling secure IoT software updates via AWS S3 presigned URLs, checksum validation, and recovery mechanisms resilient to device reboots and unreliable networks.
- Created a ReactJS dashboard for managing device groups and tracking installations, streamlining updates across thousands of devices.